

$$\Rightarrow a + c = 7 - 2 = 5 \quad \dots(i)$$

and geometric mean $= (a^2 \times b^2 \times c^2)^{1/6}$

$$\Rightarrow 2 = (abc)^{1/3} \Rightarrow abc = 8 \Rightarrow ac = \frac{8}{b} = 4 \quad [\because b = 2] \dots(ii)$$

$\therefore$ From Eqs. (i) and (ii), we get, $a = 1$, $c = 4$ and $b = 2$

50. (b) Required sum $= 2(a)^2 + 2(b)^2 + 2(c)^2$
 $= 2(1)^2 + 2(2)^2 + 2(4)^2 = 2 + 8 + 32 = 42$

51. (d) The value of c is 4.

52. (d) $\therefore$ Mode = 3 (median) - 2 (mean)
 $= 3(2) - 2\left(\frac{7}{3}\right) = 6 - \frac{14}{3} = \frac{18-14}{3} = \frac{4}{3}$

53. (c) Percentage of wheat production $= \frac{70^\circ}{360^\circ} \times 100\% = 19.44\%$

Hence, the percentage of wheat production is 19.44%.

54. (c) If $40^\circ = 9000 \text{ kg} \Rightarrow 70^\circ = \frac{9000 \times 70}{40} = 15750 \text{ kg}$

Hence, the wheat production is 15750 kg.

55. (b) $360^\circ = 180000 \text{ kg}$

Difference in Sugar and wheat angle $= (70^\circ - 40^\circ) = 30^\circ$

$\therefore$ Required production $= \frac{180000 \times 30}{360} = 15000 \text{ kg}$

Hence, the difference in sugar and wheat production is 15000 kg.

Solutions (Q. Nos. 56-58)

Class interval	f _i	cf	x _i	f _i x _i
1-4	6	6	2.5	15.0
4-7	30	36	5.5	165.0
7-10	40	76	8.5	340.0
10-13	16	92	11.5	184.0
13-16	4	96	14.5	58.0
16-19	4	100	17.5	70.0
Total	N = 100			832.0

56. (c) We have, $\frac{N}{2} = \frac{100}{2} = 50$, so cf just greater than 50 is 76 which lies in class 7-10.

$\therefore$ Median class = 7-10, $l = 7$, $f = 40$, $c = 36$, $h = 3$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - c}{f} \right) \times h = 7 + \left(\frac{50 - 36}{40} \right) \times 3 = 7 + \frac{42}{40} = 8.05$$

57. (b) The maximum frequency is 40, which lies in the class = 7-10

$\therefore$ Modal class = 7-10 $\Rightarrow l = 7$, $f_1 = 40$, $f_0 = 30$, $f_2 = 16$, $h = 3$

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

$$= 7 + \left(\frac{40 - 30}{2 \times 40 - 30 - 16} \right) \times 3 = 7 + \frac{30}{34} = 7 + 0.88 = 7.88$$

58. (d) Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{832}{100} = 8.32$

$\therefore$ Mean + Mode - 2 Median $= 8.32 + 7.88 - 2 \times 8.05$
 $= 16.2 - 16.1 = 0.1$

59. (b) In an increasing order arrangements of observations, the median will not be affected when the smallest and the largest observations are removed.

60. (c) Primary data are those data which is collected by investigator himself, while secondary data are those which is collected by other persons.

Hence, both statements are correct.

61. (b) In pie diagrams, all the items are converted into angles.

62. (b) Given that, the arithmetic mean of the above distribution = 3.1

$$\Rightarrow \frac{\sum_{i=1}^5 x_i f_i}{\sum_{i=1}^5 f_i} = 3.1 \Rightarrow \frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + x_4 f_4 + x_5 f_5}{f_1 + f_2 + f_3 + f_4 + f_5} = 3.1$$

$$\Rightarrow \frac{(1)(3) + (2)(f) + (3)(6) + (4)(5) + (5)(3)}{3 + f + 6 + 5 + 3} = 3.1$$

$$\Rightarrow \frac{3 + 2f + 18 + 20 + 15}{17 + f} = 3.1 \Rightarrow 2f + 56 = 3.1f + 52.7$$

$$\Rightarrow 1.1f = 3.3 \Rightarrow f = 3$$

Hence, the value of f is 3.

63. (a) The weight of a person is continuous variable because the weight of the persons can be divide into different but certain range of intervals. But educational qualification of the person does not show the attributes. Hence, statement I is true and statement II is false.

64. (b) Let 100 value be $x_1, x_2, x_3, \dots, x_{100}$

$$\therefore \text{Mean} = \frac{\text{Sum of values}}{\text{Total number of values}}, 45 = \frac{x_1 + x_2 + x_3 + \dots + x_{100}}{100}$$

$$\Rightarrow x_1 + x_2 + x_3 + \dots + x_{100} = 4500 \quad \dots(i)$$

Now, according to the question

$\therefore$ Sum of new value

$$= (x_1 + 15 + x_2 + 15 + \dots + x_{40} + 5) + x_{41} - 5 + x_{42} - 5 + \dots + x_{100} - 5$$

$$\therefore \text{New mean} = \frac{\text{Sum of new value}}{\text{Number of values}}$$

$$\frac{(x_1 + x_2 + x_3 + \dots + x_{40}) + 40 \times 15 + (x_{41} + x_{42} + \dots + x_{100}) - 5 \times 60}{100}$$

$$= \frac{(x_1 + x_2 + \dots + x_{100}) + 600 - 300}{100} = \frac{4500 + 300}{100} \quad [\text{from Eq. (i)}]$$

$$= \frac{4800}{100} = 48$$

Hence, the new mean is 48.

65. (a) Given numbers are 10, 7, 8, 5, 6, 8, 5, 8 and 6.

First we arrange these numbers in ascending order.

$$5, 5, 6, 6, 7, 8, 8, 8, 10$$

Total term, $n = 9$ (odd)

Now,

(i) Mean $= \frac{5 + 5 + 6 + 6 + 7 + 8 + 8 + 8 + 10}{9} = \frac{63}{9} = 7$

(ii) Median $= \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term} = \left(\frac{9+1}{2} \right)^{\text{th}} \text{ term} = 5^{\text{th}} \text{ term} = 7$

(iii) Mode = 8 (higher frequency term)

$\therefore$ Mean = Median

66. (c) In histogram the width of the bars is proportional to class interval.

67. (b) A frequency polygon can be drawn joining the mid-points of the respective tops of the rectangle of a histogram in the case of equal class intervals.

68. (c) A frequency distribution is a comprehensive way to classify raw data of a quantitative variable. It shows how different values of a variable are distributed in different classes along with their corresponding class frequencies. It is an equivalent representation of original data.

69. (d)

CI	x	f	$f \times x$
0-10	5	x	$5x$
10-20	15	$x + 8$	$15(x + 8)$
20-30	25	$x - 2$	$25(x - 2)$
30-40	35	$x - 4$	$35(x - 4)$
Total		$\Sigma f = 4x + 2$	$\Sigma fx = 80x - 70$

$$\therefore \text{Mean} = \frac{\Sigma fx}{\Sigma f} \Rightarrow 17.8 = \frac{80x - 70}{4x + 2}$$

$$\Rightarrow 17.8(4x + 2) = 80x - 70 \Rightarrow 71.2x + 35.6 = 80x - 70$$

$$\Rightarrow 8.8x = 105.6 \Rightarrow x = \frac{105.6}{8.8} = 12$$

70. (b)

Class interval (ci)	0-10	10-20	20-30	30-40
Frequencies (f)	12	20	10	8
Cumulative frequency (cf)	12	32	42	50

Here, $N = 50 \Rightarrow \frac{N}{2} = 25$

The frequency just greater than 25 is 32.

So, median class is 10-20.

71. (b) Here, modal class is 10-20, because it has maximum frequency i.e. 20.

72. (c) Given that, mean of 7 observations = 7

$$\Rightarrow \text{New mean} = \text{Previous mean} + \frac{\text{Number added to each term}}{7}$$

$$= 7 + 2 = 9$$

73. (a) Given that, number of male employees (M) = 45

Number of female employees (F) = 15

Mean salary of male employees ($\bar{x}_M$) = ₹ 5000

Total number of employees = $(M + F) = 45 + 15 = 60$

Mean salary of total employees ($\bar{x}_{MF}$) = ₹ 4800

Let mean salary of female employees be $\bar{x}_F$.

$$\bar{x}_{MF} = \frac{M\bar{x}_M + F\bar{x}_F}{(M + F)} \Rightarrow 4800 = \frac{45 \times 5000 + 15 \times \bar{x}_F}{60}$$

$$\Rightarrow 4800 \times 60 - 45 \times 5000 = 15 \times \bar{x}_F$$

$$\therefore \bar{x}_F = \frac{4800 \times 60 - 45 \times 5000}{15} = \frac{300(16 \times 4 - 50)}{15} = 300 \times 14 = ₹ 4200$$

Hence, the mean salary of female employees is ₹ 4200.

74. (a) Median = $l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$

$$\Rightarrow 14.4 = 12 + \left(\frac{\frac{20}{2} - (4 + x)}{5} \right) \times 6 = 12 + \frac{10 - 4 - x}{5} \times 6$$

$$\Rightarrow 14.4 - 12 = \frac{6 - x}{5} \times 6 \Rightarrow 2.4 = \frac{36 - 6x}{5}$$

$$\Rightarrow 12 = 36 - 6x \Rightarrow 6x = 24 \Rightarrow x = 4$$

75. (b) We have, $4 + x + 5 + y + 1 = 20 \Rightarrow x + y = 10$

$$\Rightarrow 4 + y = 10 \quad (\because x = 4) \Rightarrow y = 6$$

$$\text{Now, } \frac{x}{y} = \frac{4}{6} = \frac{2}{3} \Rightarrow 3x = 2y$$

76. (a) $\therefore \text{Mean} = \frac{\text{Sum of the observations}}{\text{Number of observations}}$

$$\Rightarrow \frac{p + q + r + s + t + u + v}{7} = m$$

$$\Rightarrow p + q + r + s + t + u + v = 7m \quad \dots(i)$$

$$\therefore (p - m) + (q - m) + (r - m) + (s - m) + (t - m)$$

$$+ (u - m) + (v - m)$$

$$= (p + q + r + s + t + u + v) - 7m \quad [\text{from Eq. (i)}]$$

$$= 7m - 7m = 0$$

77. (d) $\therefore \text{Average} = \frac{\text{Sum of the observations}}{\text{Number of observations}}$

$$\Rightarrow \frac{u + v + w + x + y + z}{6} = 10 \quad (\because \text{average} = 10)$$

$$\Rightarrow u + v + w + x + y + z = 10 \times 6 = 60 \quad \dots(ii)$$

Now, sum of new observations

$$u + 10 + v + 20 + w + 30 + x + 40 + y + 50 + z + 60$$

$$= (u + v + w + x + y + z) + 10 + 20 + 30 + 40 + 50 + 60$$

$$= 60 + 10 + 20 + 30 + 40 + 50 + 60 = 270 \quad [\text{from Eq. (i)}]$$

Now, average of $u + 10, v + 20, w + 30, x + 40, y + 50$

$$\text{and } z + 60 = \frac{270}{6} = 45$$

78. (a)

Class interval	x_i	f_i	$x_i f_i$
11-13	12	3	36
13-15	14	6	84
15-17	16	9	144
17-19	18	13	234
19-21	20	f	$20f$
21-23	22	5	110
23-25	24	4	96
Total		$\Sigma f_i = 40 + f$	$\Sigma x_i f_i = 704 + 20f$

Here, $\Sigma f_i = 40 + f$ and $\Sigma x_i f_i = 704 + 20f$ and mean, $\bar{x} = 18$

$$\therefore \text{Mean, } \bar{x} = \frac{\Sigma x_i f_i}{\Sigma f_i} \Rightarrow 18 = \frac{704 + 20f}{40 + f}$$

$$\Rightarrow 704 + 20f = 720 + 18f$$

$$\Rightarrow 20f - 18f = 720 - 704 \Rightarrow 2f = 16 \Rightarrow f = 8$$

Hence, the value of f is 8.

79. (a) Frequency polygon is formed by joining the mid-points of histogram. The original frequency distribution can be reconstructed from frequency polygon. Frequency polygon does not touch the X -axis in its extreme right and extreme left.

Hence, only statement I is true.

80. (c) I. Mean of all observations = $\frac{2 \times 4 + 7 \times 4}{8} = \frac{8 + 28}{8} = 4.5$

For median, first we arrange the given observations in ascending order = 2, 2, 2, 2, 7, 7, 7, 7, $n = 8$ (even)

$$\therefore \text{Median} = \frac{\frac{n}{2} \text{th term} + \left(\frac{n}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\frac{8}{2} \text{th term} + \left(\frac{8}{2} + 1\right) \text{th term}}{2} = \frac{4\text{th} + 5\text{th}}{2} = \frac{2 + 7}{2} = 4.5$$

II. Mode is both 2 and 7, since frequency of occurrence is same i.e. maximum frequency.

81. (c) A list of annual incomes of the members of a club is a good example of statistical data.

82. (b)

Income less than	Class interval	Number of persons	Frequency
1500	1250-1500	100	20
1250	1000-1250	80	10
1000	750-1000	70	15
750	500-750	55	23
500	250-500	32	20
250	0-250	12	12

Here, maximum frequency is 23.

So, the modal class is 500-750.

83. (c) Geometric mean of $(x_1, x_2, x_3, \dots, x_n) = (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n} = X$
and geometric mean of $(y_1, y_2, y_3, \dots, y_n)$
 $= (y_1 \cdot y_2 \cdot \dots \cdot y_n)^{1/n} = Y$
 $\therefore$ Geometric mean of $(x_1 y_1, x_2 y_2, \dots, x_n y_n)$
 $= (x_1 y_1 \cdot x_2 y_2 \cdot \dots \cdot x_n y_n)^{1/n}$
 $= (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n} \times (y_1 \cdot y_2 \cdot \dots \cdot y_n)^{1/n} = XY$
 Geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \dots, \frac{x_n}{y_n}\right) = \left(\frac{x_1}{y_1} \cdot \frac{x_2}{y_2} \cdot \dots \cdot \frac{x_n}{y_n}\right)^{1/n}$
 $= \frac{(x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n}}{(y_1 \cdot y_2 \cdot \dots \cdot y_n)^{1/n}} = \frac{X}{Y}$

84. (c) Arithmetic mean uses all the data and is always uniquely defined. Median uses only one or two numbers from the data and may not be unique. $\left(\frac{n+1}{2}\right)$ th term for odd n and half the sum of $\left(\frac{n}{2} + 1\right)$ th and $\frac{n}{2}$ th terms for even n .
85. (a) If x_i 's are the mid-points of the class intervals of grouped data, f_i 's are the corresponding frequencies and $\bar{x}$ is the mean, then $\sum f_i (x_i - \bar{x}) = 0$
86. (d) When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the areas proportional to the frequencies of the respective class intervals, the graph so obtained is called histogram.
87. (c) Both statements I and II are suitable for representation of a pie diagram.
88. (a) Here, statement I is correct but statement II is not correct.
89. (c) The class which has maximum frequency is known as modal class.
90. (c) Given observations in ascending order are 6, 14, 15, 17, $x + 1$, $2x - 13$, 30, 32, 34 and 43.

Here, $n = 10$

(even)

$$\therefore \text{Median} = \frac{\text{Value of } \left(\frac{n}{2}\right) \text{th term} + \text{Value of } \left(\frac{n}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\text{Value of } \left(\frac{10}{2}\right) \text{th term} + \text{Value of } \left(\frac{10}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\text{Value of 5th term} + \text{Value of 6th term}}{2}$$

$$= \frac{x + 1 + 2x - 13}{2} = \frac{3x - 12}{2}$$

But given, median = 24 $\Rightarrow \frac{3x - 12}{2} = 24$

$\Rightarrow 3x - 12 = 24 \times 2 = 48 \Rightarrow 3x = 48 + 12$

$\Rightarrow 3x = 60 \Rightarrow x = 20$

Hence, the value of x is 20.

Solutions (Q. Nos. 91-94)

Class interval	Frequency (f)	Class mark (x)	f(x)	cf
0-10	4	5	20	4
10-20	5	15	75	9
20-30	7	25	175	16
30-40	10	35	350	26
40-50	12	45	540	38
50-60	8	55	440	46
60-70	4	65	260	50
Total	50		1860	

91. (a) Mean = $\frac{\sum fx}{\sum f} = \frac{1860}{50} = 37.2$

Hence, the value of mean is 37.2.

92. (b) Here, $N = 50$, Now, $\frac{N}{2} = \frac{50}{2} = 25$

which lies in the cumulative frequency corresponding class interval for cf 26 is 30-40.

93. (c) From the table, $l = 30$, $f = 10$, $C = 16$ and $h = 10$

$$\therefore \text{Median} = l + \frac{\left(\frac{N}{2} - C\right)}{f} \times h$$

$$= 30 + \frac{(25 - 16)}{10} \times 10 = 30 + 9 = 39$$

94. (d) Modal class of the given data is 40-50, because it has largest frequency among the given classes of the data i.e. 12.

Here, $l = 40$, $f_1 = 12$, $f_0 = 10$, $f_2 = 8$ and $h = 10$

$$\therefore \text{Mode} = l + \left\{ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right\} \times h = 40 + \left\{ \frac{12 - 10}{2 \times 12 - 10 - 8} \right\} \times 10$$

$$= 40 + \frac{2 \times 10}{24 - 18} = 40 + \frac{20}{6} = 40 + 3.33 = 43.33$$

Hence, the mode of given data is 43.33.

95. (b) Given, total number of votes = 100000

Party E get votes = $100000 - (36000 + 24000 + 18000 + 7000)$
 $= 100000 - 85000 = 15000$

$\therefore$ Angle allocated for party E = $\frac{15000}{100000} \times 360^\circ = 54^\circ$